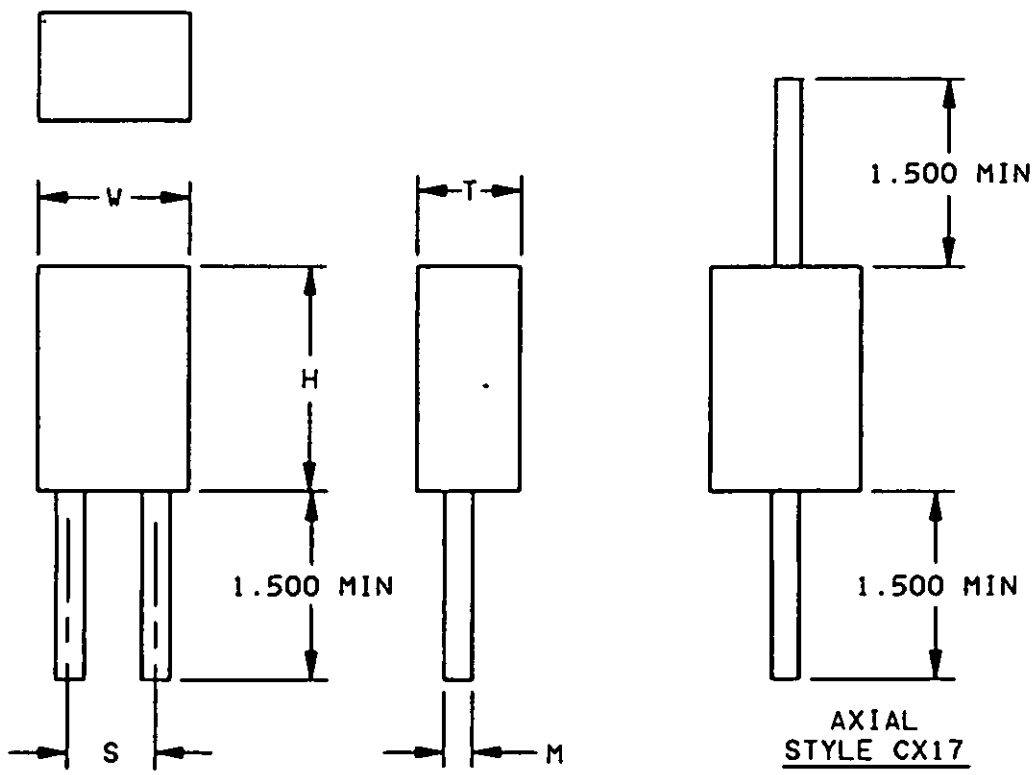
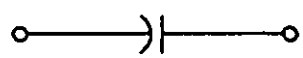
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, NON-POLAR, NONHERMETICALLY SEALED, STYLES CX07 AND CX17

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-49137.





CIRCUIT DIAGRAM

FIGURE 1. Styles CX07 and CX17 capacitors.

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FSC 5910

Case size	Dimensions							
	T		W		H		M	S
	Min	Max	Min	Max	Min	Max	±.002	
	CX07							
A	.095	.125	.095	.125	.150	.180	.010	.100 ±.020
B	.150	.180	.110	.140	.210	.240	.010	.100 ±.020
C	.190	.220	.150	.180	.285	.315	.016	.150 ±.025
D	.200	.230	.200	.230	.305	.335	.016	.200 ±.025
	CX17							
E	.095	.125	.095	.125	.190	.220	.010	.100 ±.020
F	.150	.180	.110	.140	.250	.280	.010	.100 ±.020
G	.190	.220	.150	.180	.340	.370	.016	.150 ±.025
H	.200	.230	.200	.230	.350	.390	.016	.200 ±.025

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.110	2.79	.210	5.33	.305	7.75
.010	.25	.125	3.18	.220	5.59	.315	7.91
.016	.38	.140	3.56	.230	5.84	.335	8.47
.020	.41	.150	3.81	.240	6.10	.340	8.64
.025	.51	.180	4.57	.250	6.35	.350	8.89
.095	2.41	.190	4.83	.280	7.11	.370	9.40
.100	2.54	.200	5.08	.285	7.24	.390	9.91
						1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Styles CX07 and CX17 capacitors - Continued.

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TABLE I. Styles CX07 and CX17 characteristics.

Type designation 1/	DC rated voltage (85°C)	Capacitance (nom)	Dissipation factor (max)	DC leakage (25°C) (max)	Case size	
					CX07	CX17
	Volts	µF	Percent	µA		
CX-7A475-	2	4.7	10	0.5	A	E
CX-7B335-	3	3.3	10	0.5	A	E
CX-7B106-	3	10.0	10	0.5	B	F
CX-7C225-	4	2.2	8	0.5	A	E
CX-7C685-	↓	6.8	↓	1.0	B	F
CX-7C226-	↓	22.0	↓	2.0	C	G
CX-7C336-	↓	33.0	↓	3.0	D	H
CX-7D155-	6	1.5	6	0.5	A	E
CX-7D475-	↓	4.7	↓	1.0	B	F
CX-7D156-	↓	15.0	↓	2.0	C	G
CX-7D226-	↓	22.0	↓	3.0	D	H
CX-7F105-	10	1.0	6	0.5	A	E
CX-7F335-	↓	3.3	↓	1.0	B	F
CX-7F106-	↓	10.0	↓	2.0	C	G
CX-7F156-	↓	15.0	↓	3.0	D	H
CX-7H684-	15	0.68	6	0.5	A	E
CX-7H685-	15	6.8	6	2.0	C	G
CX-7H106-	15	10.0	6	3.0	D	H
CX-7J474-	20	0.47	6	0.5	A	E
CX-7J155-	↓	1.5	↓	1.0	B	F
CX-7J225-	↓	2.2	↓	1.0	B	F
CX-7J475-	↓	4.7	↓	2.0	C	G
CX-7J685-	↓	6.8	↓	3.0	D	H
CX-7K334-	25	0.33	6	0.5	A	E
CX-7K105-	↓	1.0	↓	1.0	B	F
CX-7K335-	↓	3.3	↓	2.0	C	G
CX-7K475-	↓	4.7	↓	3.0	D	H
CX-7M104-	35	0.10	6	0.5	A	E
CX-7M154-	↓	0.15	↓	↓	↓	↓
CX-7M224-	↓	0.22	↓	↓	↓	↓
CX-7M334-	↓	0.33	↓	↓	↓	↓
CX-7M474-	↓	0.47	↓	1.0	B	F
CX-7M684-	↓	0.68	↓	1.0	B	F
CX-7M105-	↓	1.0	↓	2.0	C	G
CX-7M155-	↓	1.5	↓	2.0	C	G
CX-7M225-	↓	2.2	↓	2.0	C	G
CX-7M335-	↓	3.3	↓	3.0	D	H

1/ Complete type designation will include additional symbols to indicate style and capacitance tolerance.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case material: Plastic encapsulated.

Terminals: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55° to +85°C.

Construction analysis: Not applicable.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

Cap. tolerance: ±10% (K) or ±20% (M).

Dissipation factor (DF): See table I.

Shock (specified pulse): Not applicable.

Terminal strength: Method 211 of MIL-STD-202.

Test condition A - 1 pound for case sizes A, B, E, and F; 2 pounds for case sizes C, D, G, and H.

Test condition C - 1/2 pound applied force - 3 bends.

Stability at low and high temperatures: In accordance with MIL-C-49137.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Within tolerance of table I value.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Within -15 percent of step 1 measured value.
	DF	-	200 percent of table I value.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	10 times 25°C limit (see table I).
	ΔCap.	-	Within +15 percent of step 1 measured value.
	DF	-	150 percent of table I value.
Step 5 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.

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Surge voltage: In accordance with MIL-C-49137.

DCL - See table I.

ΔCap. - Within 10 percent of initial measured value.

DF - See table I.

Life: Method 108 of MIL-STD-202, test condition D (1,000 hours):

At 25°C:

DCL - 200 percent of table I value.

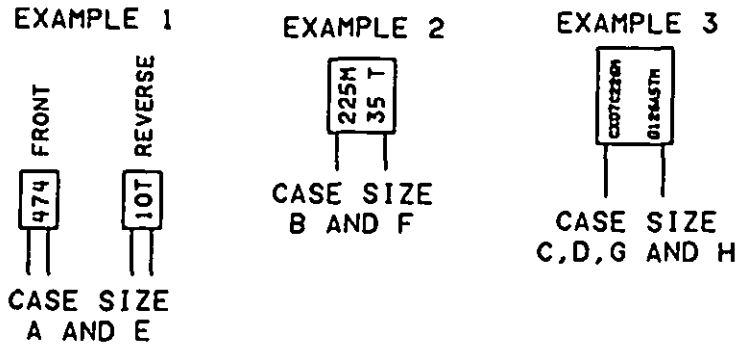
ΔCap. - Within 10 percent of initial measured value.

DF - See table I.

Marking: In accordance with MIL-C-49137 and the following:
Case sizes A and E shall be marked as shown in example 1. The front shall be marked with the capacitance value coded in picofarads and the reverse shall be marked with the dc rated voltage and the manufacturer's trademark.

Case sizes B and F shall be marked as shown in example 2 (see MIL-C-49137).

Case sizes C, D, G, and H shall be marked as shown in example 3. The first line shall contain the type designation and the second line shall contain the date code (year and week), lot code, and manufacturer's trademark.



Voltage groups (85°C rated voltage):
I----------2 to 10V inclusive
II----------15 to 35V inclusive

Custodian:
Army - ER

Review activity:
DLA - ES

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5910-A375)

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